# MILITARY SPECIFICATION

## VOLTAGE REGULATOR SUBASSEMBLY: 10525760

## 1. SCOPE

1.1 This specification covers one type of voltage regulator subassembly which contains the regulator and driver circuits for the +143 volt, +78 volt, +66 volt, +1.25 volt, and -18 volt regulated power supplies used for field artillery applications.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-F-13926 | Fire Control Materiel, General Specification Governing the Manufacture and Inspection of |
| MIL-P-14232 | Parts, Equipment and Tools for Ordnance Materiel: Packaging of |
| MIL-I-45607 | Inspection Equipment, Supply and Maintenance of |
| MIL-C-46357 | Computer, Gun Direction: M18; Manufacturing Processess and Inspection of |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | Quality Assurance Terms and Definitions |
| MIL-STD-202 | Test Methods for Electronic and Electrical Compoent Parts |

DRAWINGS

U.S. Army Munitions Command

| | |
|---|---|
| F10525760 | Voltage Regulator Subassembly |

PACKAGING DATA SHEET

| | |
|---|---|
| M14002 | Data Sheet, Master Packaging |

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(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 The voltage regulator shall be manufactured in accordance with Drawing F10525760 and drawings pertaining thereto.

3.2 General specifications.- The voltage regulator shall meet the following requirements of MIL-F-13926, and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship

3.2.2 MIL-C-46357

a. Etched circuit assembly
b. Marking
c. Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The voltage regulator shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the voltage regulator for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the voltage regulator shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The voltage regulator shall withstand 3 shock impulses in the vertical plane, relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3 Temperatures.

3.3.3.1 Operating.- The voltage regulator shall meet the requirements of 3.4.4 while exposed to temperatures of -40°F and +135°F.



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3.3.3.2 Storage.- The voltage regulator shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

3.3.4 Humidity.

3.3.4.1 Operating.- The voltage regulator shall meet the requirements of 3.4.4 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 135°F.

3.3.4.2 Non-operating.- The voltage regulator shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The voltage regulator shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The voltage regulator shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a fine mist of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Performance.- Unless otherwise specified the voltage regulator shall meet the performance requirements of this specification at ambient temperatures between 60°and 90°F; a barometric pressure of 28 to 32 inches of mercury; and a RH up to 90%. Air at ambient temperatures shall be circulated across the components at the rate of at least 300 feet per minute.

3.4.1 Wiring.- Resistors R9, R10, R22, R27, R28 R71 and R72 shall be selected and assembled on the voltage regulator prior to subsequent requirements. Wiring and connections shall be in accordance with electric schematic Drawing F10525763.

3.4.2 Resistors.- After final assembly onto the voltage regulator, resistors which are not functionally part of the circuits referenced in 3.4.4 shall possess characteristics within tolerance limits specified on the appropriate drawings.

3.4.3 Thermistors.- After final assembly onto the voltage regulator, thermistors which are not functionally part of the circuit of 3.4.4 shall possess characteristics within tolerance limits specified on the appropriate drawings.

3.4.4 Circuit outputs.- The circuits of the voltage regulator are part of a regulated power supply and when connected to the appropriate external circuitry shall have outputs as shown in Table I; when energized at input voltages corresponding to high and low ac line voltage; and when supplying maximum and minimum load currents or overload currents.

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TABLE I

Output Voltages and Load Currents

| Test Number | Input Power Supply Voltage Settings (±2 percent) Power Supply No. 3 (Volts) | Power Supply No. 5 (Volts) | Power Supply No. 7 (Volts) | Load Current Settings (±3 percent) +1.25V load (Amperes) | -18V load (Amperes) | Measured Output Voltages +1.25V loaded (Volts) | -18V loaded (Volts) | Maximum Measured Output Ripple* +1.25V load (Volts) | -18V load (Volts) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 14 | 29 | 41 | 1.6 | 4.5 | 1.25 ± 0.013 | -18 ± .20 | 0.1 | 0.5 |
| 2 | 11 | 23 | 31 | 1.88 | 3.2 | 1.25 ± .038 | -18 ± .54 | 0.1 | 0.5 |
| 3 | 11 | 23 | 31 | 1.25 | 5.6 | 1.25 ± .038 | -18 ± .54 | 0.1 | 0.5 |
| 4 | 17 | 34 | 49 | 1.88 | 3.2 | 1.25 ± .038 | -18 ± .54 | 0.1 | 0.5 |
| 5 | 17 | 34 | 49 | 1.25 | 5.6 | 1.25 ± .038 | -18 ± .54 | 0.1 | 0.5 |
| 6 | 14 | 29 | 41 | 1.6 | 4.5 | 1.313 ± .030 | -18.9 ± .38 | | |
| 7 | 14 | 29 | 41 | 1.6 | 4.5 | 1.188 ± .030 | -17.1 ± .38 | | |
| 8 | 14 | 29 | 41 | 1.6 | 4.5 | 1.25 ± 0.50 | -18.0 ± 2.0 | | |
| 9 | 11 | 23 | 31 | 1.88 | 3.2 | 1.25 ± .062 | -18 ± .9 | 0.1 | 0.5 |
| 10 | 11 | 23 | 31 | 1.25 | 5.6 | 1.25 ± .062 | -18 ± .9 | 0.1 | 0.5 |
| 11 | 17 | 34 | 49 | 1.88 | 3.2 | 1.25 ± .062 | -18 ± .9 | 0.1 | 0.5 |
| 12 | 17 | 34 | 49 | 1.25 | 5.6 | 1.25 ± .062 | -18 ± .9 | 0.1 | 0.5 |

*NOTE: Ripple is defined as the peak-to-peak value of any alternating component of the output voltage.

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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables II, III, and IV and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table II shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60 to 90 F).

TABLE II

(100 percent inspection)

| Characteristic | Requirement | Test Procedures |
|---|---|---|
| 101. Wiring | 3.4.1 | 4.5.2.1 |
| 102. Resistors | 3.4.2 | 4.5.2.2 |
| 103. Thermistors | 3.4.3 | 4.5.2.3 |
| 104. Circuit outputs | 3.4.4 | 4.5.2.4 |



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4.3 Standard sampling.

4.3.1 General.- One unit shall be selected at random by a Government representative as a standard sample from each 100 produced. The sample shall meet the requirements and tests in Table III.

TABLE III

| Characteristics | REQUIREMENT | TEST PROCEDURES |
|---|---|---|
| 301. Fabrication | 3.1 | Applicable drawings - visual |
| 302. General specifications | 3.2 | MIL-F-13926 and MIL-C-46357 - visual |

4.3.2 Environmental.- Three units shall be selected at random by a Government representative as a sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall have met the requirements and tests in Table II and shall then meet the requirements and tests in Table IV.

TABLE IV

| Characteristics | REQUIREMENT | TEST PROCEDURES |
|---|---|---|
| 303. Vibration | 3.3.1 | 4.5.1.1 |
| 304. Shock | 3.3.2 | 4.5.1.2 |
| 305. Humidity-Operating | 3.3.4.1, 3.4.4 | 4.5.1.3.2, 4.5.2.4 |
| 306. Humidity-Non-operating | 3.3.4.2 | 4.5.1.3.1 |
| 307. Salt spray | 3.3.6 | 4.5.1.4 |
| 308. Operating temperature | 3.3.3.1, 3.4.4 | 4.5.1.5, 4.5.2.5 |
| 309. Storage temperature | 3.3.3.2 | 4.5.1.5 |

4.3.3 Failure of sample.- Should any one item of a standard sampling fail to meet the specified test requirements, acceptance of the represented production lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.



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4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

| Item Number | Description |
|---|---|
| 1. Voltage measuring device | 0-50 Vdc, input resistance 10 megohms or greater, accuracy of ± .5% or better. |
| 2. Resistance measuring device | 0-200 ohms, accuracy of ± .5% of scale reading. |
| 3. Current measuring device | 0-10 amperes, accuracy of ± .1% full scale reading. |
| 4. Oscilloscope | Rise time of scope plus measuring probe; 50 nano seconds per centimeter, input resistance 8 megohms and 25 pico-farads or greater. |

4.4.1.2 In addition to the test equipment specified above, utilize the power supplies resistors, signal sources, transistors, capacitors, potentiometers, diodes, and transformers designated in Tables I, V, VI, VII, and Figures 1 to 6.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on adevice with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plu or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the unit rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock.- This test shall be conducted in accordance with the conditions of 3.3.2 and Procedure II of the shock test defined in MIL-F-13926 except that impact shocks shall not exceed 5 G's. The unit shall be subsequently inspected for physical damage or defects in compliance with 3.3.2. The unit shall then be inspected as specified in 4.5.1.3 prior to determining compliance with the requirements of 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dired and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.



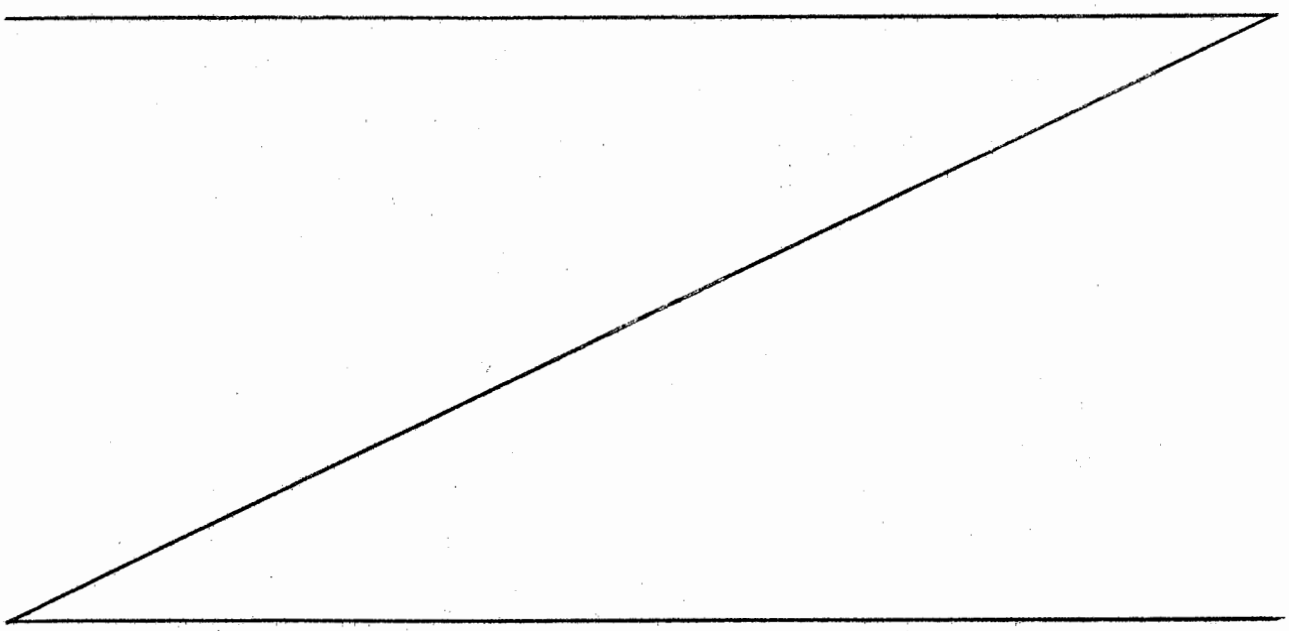
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4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 95 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.2.4 to determine compliance with 3.3.4.1.

4.5.1.4 Salt spray.- The unit shall be subjected to the conditions in 3.3.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.2.2 to 4.5.2.4.2.2 inclusive for compliance with 3.3.6.

4.5.1.5 Temperature (Operating and storage).- After being subjected to all environmental tests specified above, the unit shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the unit shall be tested in accordance with 4.5.2.5 to determine compliance with the requirements of 3.4.4 as required by 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.2.2 to 4.5.2.4.2.2 inclusive to determine compliance with 3.3.3.2.

4.5.1.6 Fungus.- To determine compliance with 3.3.5, one of the first 5 units produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.





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# TABLE V(a)

## RESISTOR SELECTION LIST*

| Nominal Value (Ohms) | Part Number |
|---|---|
| 1.7 | 10526597 |
| 3.4 | 10526701 |
| 5. | 10526702 |
| 6. | 10526703 |
| 6.7 | 10526704 |
| 8.4 | 10526705 |
| 10. | 10526706 |
| 11.7 | 10526707 |
| 13.4 | 10526708 |
| 15. | 10526709 |
| 16.7 | 10526710 |
| 18.4 | 10526711 |
| 20. | 10526712 |
| 21.7 | 10526713 |
| 23.4 | 10526714 |
| 25. | 10526715 |
| 26.7 | 10526716 |
| 28.4 | 10526717 |
| 30. | 10526718 |
| 35. | 10526719 |
| 40. | 10526720 |
| 45. | 10526721 |
| 50. | 10526722 |
| 55. | 10526723 |
| 60. | 10526724 |
| 65. | 10526725 |
| 70. | 10526726 |
| 75. | 10526727 |
| 80. | 10526728 |
| 85. | 10526729 |
| 90. | 10526730 |
| 95. | 10526731 |
| 100. | 10526732 |
| 105. | 10526733 |
| 6.K | 10526651 |
| 6.5K | 10526652 |
| 7.K | 10526653 |
| 7.5K | 10526654 |
| 8.K | 10526655 |
| 8.5K | 10526656 |
| 9.K | 10526657 |
| 9.5K | 10526658 |
| 10.K | 10526659 |



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## TABLE V(b)

## RESISTOR SELECTION LIST

### Potentiometer Arm Resistance Connections

| Resistor No. | From Terminal | To Terminal |
|---|---|---|
| R9 | 34A | 32B |
| R10 | 31B | 32B |
| R22 | 1B | 5A |
| R27 | 23B | 29B |
| R28 | 58B | 29B |
| R71 | 10A | 15B |
| R72 | 18B | 15B |

NOTE: *R22 varies between 6K and 10K ohms
R27 and R28 varies between 1.7 and 28.4 ohms
R71 and R72 varies between 5 and 65 ohms
R9 and R10 varies between 5 and 105 ohms

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## TABLE VI

### External Circuitry Component Values

| Component Number | Part Number |
|---|---|
| *Q1, Q2 | 10525749-1 |
| **Q3, Q4, Q5, Q7, Q8 | 10525764 |
| Q6 | 10525749-2 |
| Q9 | 10525779 |
| ***CR1, CR5 | 10525738 |
| CR2 | 10525737 |
| ***CR3, CR4 | 10525769 |
| C1, C4 | 10525817 |
| C2, C3 | 10525754 |
| C5, C8 | 10525820 |
| C6 | 10525829 |
| C9 | 10525824 |
| C7 | 10525810 |
| L1 | 10525920 |
| T1 | 10525744 |
| R5 | 10525778-6 |
| R6 | 10525778-5 |
| R7 | 10526649 |
| R8 | MS35043-87 |
| R9 | 10526660 |
| R16 | 10526732 |
| R18 | 10525778-8 |
| R19 | 10525778-9 |

(b) Potentiometers and Resistors

| Component Number | Component Type | Component Value (ohms) | Tolerance (percent) | Wattage (watts) |
|---|---|---|---|---|
| R1 | Potentiometer | 30 | 0.5 | 0.5 |
| R2 | Potentiometer | 70 | 0.5 | 0.5 |
| R3 | Potentiometer | 110 | 0.5 | 0.5 |
| R4 | Variable Resistance | 6K to 10K inclusive | 0.5 | 0.5 |
| R10 | Fixed Resistor | 5.5K | 1 | 2 |
| R11 | Fixed Resistor | 2K | 1 | 10 |
| R12 | Fixed Resistor | 7.5K | 1 | 2 |
| R13 | Fixed Resistor | 8K | 1 | 5 |
| R14 | Fixed Resistor | 25K | 1 | 2 |
| R15 | Fixed Resistor | 82.5K | 1 | 1/8 |

*These transistors shall be mounted on heat dissipator Part No. 10525739.

**These transistors shall be mounted on the transistor mounting bracket. (Part No. 10525736)

***This diode shall be mounted on the diode mounting bracket Part No. 10525776.

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## TABLE VII

### D. C. Power Supplies

| Power Supply No. | Nominal Voltage | Voltage Tolerance (percent) | Maximum RMS Ripple (m volts) | Current Requirement (amperes) | Maximum Output Impedance (ohms) | Terminal Connections Positive Terminal | Negative Terminal |
|---|---|---|---|---|---|---|---|
| 1 | 6.0 | ± 2 | 10 | 0.15 | 1 | Load ground | 6B |
| 2 | 6.0 | ± 2 | 5 | 0.15 | 1 | 11B | Load ground |
| 3 | *10 to 20 | ± 2 | 500 | 2.0 | 0.5 | 11A, 12A, 13A, 14A, 17A, 18A, 20A, 22A, 23A, 24A, 25A, 26A, 28A, 29A, 19A | 35B, 36B, 37B, 38B |
| 4 | 15 | ± 2 | 5 | 0.1 | 1 | 61B | Load ground |
| 5 | *20 to 35 | ± 2 | 100 | 0.2 | 1 | Load ground | 63A |
| 6 | 35 | ± 2 | 5 | 0.1 | 1 | 5A | Load ground |
| 7 | *30 to 50 | ± 2 | 500 | 12.0 | 0.1 | (See Fig. #2) | (See Fig. #2) |
| 8 | *130 to 170 | ± 1 | 100 | 0.2 | 5 | 49A | 65A |
| 9 | 20 | ± 2 | 500 | 1.5 | 1 | (See Fig. #3) | Load ground |
| 10 | 12 | ± 2 | 20 | 0.2 | 1 | Load ground | 62B |

* Power supplies shall be continuously variable throughout the voltages specified.

NOTE: All power supplies will be electrically isolated from each other except as designated under terminal connections. See Figure 2 for load ground.

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4.5.2 Performance.

4.5.2.1 Wiring.- To determine compliance with 3.4.1 the voltage regulator shall be visually inspected to insure that all components are physically present and conform to Ordnance Drawing No. 10525763. If resistors R9, R10, R22, R27, R28, R71, and R72 are missing, they shall be selected and installed using the procedure indicated in 4.5.2.1.1 to 4.5.2.1.1.4 inclusive. If the voltage regulator conforms to Ordnance Drawing No. 10525763, it shall be electrically tested as indicated in 4.5.2.2 to 4.5.2.4.2.2 inclusive.

4.5.2.1.1 Selection of resistors.- The resistors listed in Table V shall be selected as indicated in this paragraph such that the voltage regulator shall meet the requirements of 4.5.2.4. Connect the required external circuitry to the board as indicated in figures 1 and 2. Set potentiometers R1 and R2 described in Table VI at approximately the middle of the resistance range. Set potentiometer R3 for minimum resistance between terminals 31B and 32B. Set variable resistor R4 for 8,000 ± 800 ohms. Adjust the +1.25-volt load for a resistance of 0.8 ± 0.08 ohms. Adjust the -18-volt load for a resistance of 3.8 ± 0.38 ohms. Set power supply No. 3 to 14 ± 0.42 volts, power supply No. 5 to -29 ± 0.87 volts, and power supply No. 7 to 41 ± 1.2 volts. Energize the power supplies of Table VII in the following sequence: Power Supply No. 1, 5, 2, 4, 6, 9, 3, 7 and 10 (Power Supply No. 8 is not used for these tests).

4.5.2.1.1.1 Oscillator frequency resistor.- Adjust variable resistance R4 such that the frequency of the voltage waveform measured from terminal 2B to terminal 3B, is as close to 1450 cycles per second as the variable resistor can be adjusted. The frequency shall be 1450 ± 30 cycles per second.

4.5.2.1.1.2 Output voltage resistors.- Adjust the +1.25-volt load to 1.6 ± 0.08 amperes. Adjust potentiometer R1 such that the output voltage across the +1.25-volt load is as close to 1.25 volts as the potentiometer resolution allows. The voltage across the +1.25-volt load shall be +1.25 ± 0.003 volts. Adjust the -18-volt load to 4.7 ± 0.24 amperes. Adjust potentiometer R2 such that the output voltage across the -18-volt load is as close to -18 volts as the potentiometer resolution allows. The voltage across the -18-volt load shall be -18 ± 0.04 volts.



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4.5.2.1.1.3 Trip circuit resistors.- Adjust the -18-volt load to 9.5 ± 0.095 amperes. Adjust potentiometer R3 until the overload circuit trips. (The circuit is in the tripped condition when the voltage across the -18-volt load is less than 1 volt for a resistive load of less than 5 ohms). De-energize all power supplies. Set the -18-volt load for a resistance of 3.8 ± 0.38 ohms. Re-energize the power supplies in the sequence of 4.5.2.1.1. Slowly increase the -18-volt load until the circuit trips. The load current when tripping occurs shall be 9.5 ± 0.285 amperes. If tolerance is exceeded, repeat the last step with various settings of R3 until circuit trips out within tolerance.

4.5.2.1.1.4 Resistor installation.- De-energize all power supplies and install the resistors selected from the appropriate column in Table V (a) to the nearest resistance arms of potentiometers R1, R2, and R3 and variable resistance R4 given in Table VI (b).

4.5.2.2 Resistors.- To determine compliance with 3.4.2 connect lead A of figure 5 to terminal 60B and lead B of figure 5 to terminal 59B. Energize power supply No. 3 and adjust power supply No. 3 until its output voltage is 15 ± 0.03 volts. The voltage from terminal 60B to terminal 59B shall be 10 ± 0.3 volts. The voltage from terminal 64A to terminal 59B shall be 5 ± 0.25 volts.

4.5.2.3 Thermistors.- To determine compliance with 3.4.3 connect lead A of figure 4 to terminal 61A and lead B of figure 4 to terminal 62A and energize power supply No. 1. The voltage shall be 0.77 ± 0.33 volts dc from terminals 61A to 62A. De-energize power supply No. 1 and connect lead A of figure 4 to terminal 1A and lead B of figure 4 to terminal 2A. Energize power supply No. 1. The voltage shall be 0.088 ± 0.04 volts dc from terminals 1A to 2A. De-energize power supply No. 1.

4.5.2.4 Circuit outputs.- Make the following connections and measurements to determine compliance with 3.4.4.

4.5.2.4.1 +1.25-volt and -18-volt regulator operation.- Connect the power supplies and external circuitry according to figure 1 and figure 2 with the potentiometers R1, R2, and R3 and variable resistor R4 removed. Adjust the loads and power supplies in accordance with 4.5.2.1.1. Energize the power supplies in the sequence given in 4.5.2.1.1.

4.5.2.4.1.1 Frequency measurement.- The frequency of the voltage waveform from terminal 2B to terminal 3B shall be 1450 ± 100 cycles per second.



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4.5.2.4.1.2 Output voltage measurements.- Output voltage measurements shall be made in accordance with tests No. 1, 2, 3, 4 and 5 of Table I. Remove all connections from terminal 32A and terminal 34B. Perform test No. 6 of Table I. Make a connection between terminals 33A and 33B and between terminals 30A and 25B. Perform test No. 7 of Table I. Remove the connections between terminals 33A and 33B and between terminals 30A and 25B. Replace the connections to terminals 34B and to terminal 32A.

4.5.2.4.1.3 Remote sensing resistors.- Remove the connection between terminal 33B and the +1.25 volt load and between terminals 30A and the -18 volt load. Remove the connection between the load ground and terminal 3A. Perform test No. 8 of Table I. Replace the connections to terminal 33B, terminal 30A, and terminal 3A.

4.5.2.4.1.4 Trip circuit.- Increase the -18-volt load until the circuit trips. The -18-volt load current when tripping occurs shall be 9.5 ± 0.85 amperes. (At no time shall the load current exceed 12 amperes). The voltage from terminals 26B to 50A shall be less than 2 volts. De-energize all power supplies.

4.5.2.4.1.5 Voltage transient.- Adjust the loads and power supplies in accordance with 4.5.2.1.1. Energize power supplies No. 1, 2, 3, 4, 5, 6, and 9. Energize power supply No. 7 and measure the voltage transient across the -18-volt load. The voltage shall not exceed 26 volts at any time. The transient shall be over in 3 seconds. De-energize all power supplies.

4.5.2.4.2 Neon nixie supply.

4.5.2.4.2.1 High voltage measurement.- Connect the circuit as shown in figure 6 with R10, R11, and R13 connected in the circuit. Energize and adjust power supply No. 8 for 158 ± 1.58 volts. The voltage from terminal 46A to terminal 65A shall be 73.4 ± 5.46 volts. The voltage from terminal 45A to terminal 65A shall be 66 ± 4.6 volts. The voltage from terminal 49A to terminal 48A shall be 12.4 ± 1.0 volts. De-energize power supply No. 8.

4.5.2.4.2.2 Low voltage measurement.- Remove R10 from the circuit. Replace R11 with R12 and replace R13 with R14. Energize power supply No. 8 and adjust the voltage to 141 ± 1.41 volts. The voltage from terminal 46A to terminal 65A shall be 73 ± 5.46 volts. The voltage from terminal 45A to terminal 65A shall be 66 ± 4.6 volts. The voltage from terminal 49A to terminal 48A shall be 12.4 ± 1.0 volts. De-energize power supply No. 8.

4.5.2.5 Extreme environmental conditions.- Utilize the set up specified in 4.5.2.4.1 to conduct the following tests.



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4.5.2.5.1 Low temperature (-40°F).- Performance at the low operating temperature specified in 3.3.3.1 shall be determined as follows to assure compliance.

4.5.2.5.1.1 Frequency measurement.- Perform the test of 4.5.2.4.1.1 except that the frequency shall be 1450 ± 190 cycles per second.

4.5.2.5.1.2 Output voltage measurements.- Output voltage measurements shall be made in accordance with tests No. 9, 10, 11, and 12 of Table I.

4.5.2.5.1.3 Trip circuit.- Perform the tests of 4.5.2.4.1.4 except that the -18-volt load current when tripping occurs shall be 10 ± 1.5 amperes.

4.5.2.5.1.4 Neon nixie supply high voltage measurement.- Perform the instructions of 4.5.2.4.2.1 except that the voltage measurements shall be 72.25 ± 8.5 volts from terminal 46A to terminal 65A, 65 ± 6.9 volts from terminal 45A to terminal 65A, and 12.4 ± 1.2 volts from terminal 49A to terminal 48A.

4.5.2.5.1.5 Neon nixie supply low voltage measurement.- Perform the instructions of 4.5.2.4.2.2 except that the voltage measurements shall be 72.25 ± 8.5 volts from terminal 46A to terminal 65A, 65 ± 6.9 volts from terminal 45A to terminal 65A, and 12.4 ± 1.2 volts from terminal 49A to terminal 48A.

4.5.2.5.2 High operating temperature - (+135°F).

4.5.2.5.2.1 Frequency measurement.- Perform the test of 4.5.2.4.1.1 except that the frequency shall be 1450 ± 190 cycles per second.

4.5.2.5.2.2 Output voltage measurements.- Perform the tests of 4.5.2.5.1.2.

4.5.2.5.2.3 Trip circuit.- Perform the tests of 4.5.2.4.1.4 except that the -18-volt load current when tripping occurs shall be 9.5 ± 1.33 amperes.

4.5.2.5.2.4 Neon nixie supply high voltage measurement.- Perform the instructions of 4.5.2.4.2.1 except that the voltage measurements shall be 72.25 ± 8.5 volts from terminal 46A to terminal 65A, 65 ± 6.9 volts from terminal 45A to terminal 65A, and 12.4 ± 1.2 volts from terminal 49A to terminal 48A.

4.5.2.5.2.5 Neon nixie supply low voltage measurement.- Perform the instructions of 4.5.2.4.2.2 except that the voltage measurements shall be 72.25 ± 8.5 volts from terminal 46A to terminal 65A, 65 ± 6.9 volts from terminal 45A to terminal 65A, and 12.4 ± 1.2 volts from terminal 49A to terminal 48A.



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## 5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with Master Packaging Data Sheet M14002.

## 6. NOTES

6.1 Intended use.

6.1.1 +1.25-volt shunt regulator.- Part of this assembly consists of the sensing and amplifier portions of a +1.25-volt shunt regulator. With the output power transistors connected to the voltage regulator and the -18-volt pulse width modulated regulator (also on this assembly) operating to supply the current, the +1.25-volt shunt regulator is designed to meet the requirements of this specification for a load of 1.27 to 1.88 amperes.

6.1.2 -18-volt pulse width modulated regulator.- Part of this assembly consists of the sensing and amplifier portions of a -18-volt pulse width modulated regulator. With the output power transistors and the filtering portions connected to the voltage regulator the -18-volt pulse width modulated regulator is designed to meet the requirements of this specification for a load of 3.2 to 5.6 amperes.

6.1.3 Nixie neon supply.- Part of this assembly consists of a nixie neon supply. The nixie neon supply, by the use of zener diodes and a +150 volt supply, will put out +138 volt, +73 volt, and +66 volt bias voltages for the nixie driver circuits.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet (see 5.1).
(e) A block of serial numbers to identify each voltage regulator.

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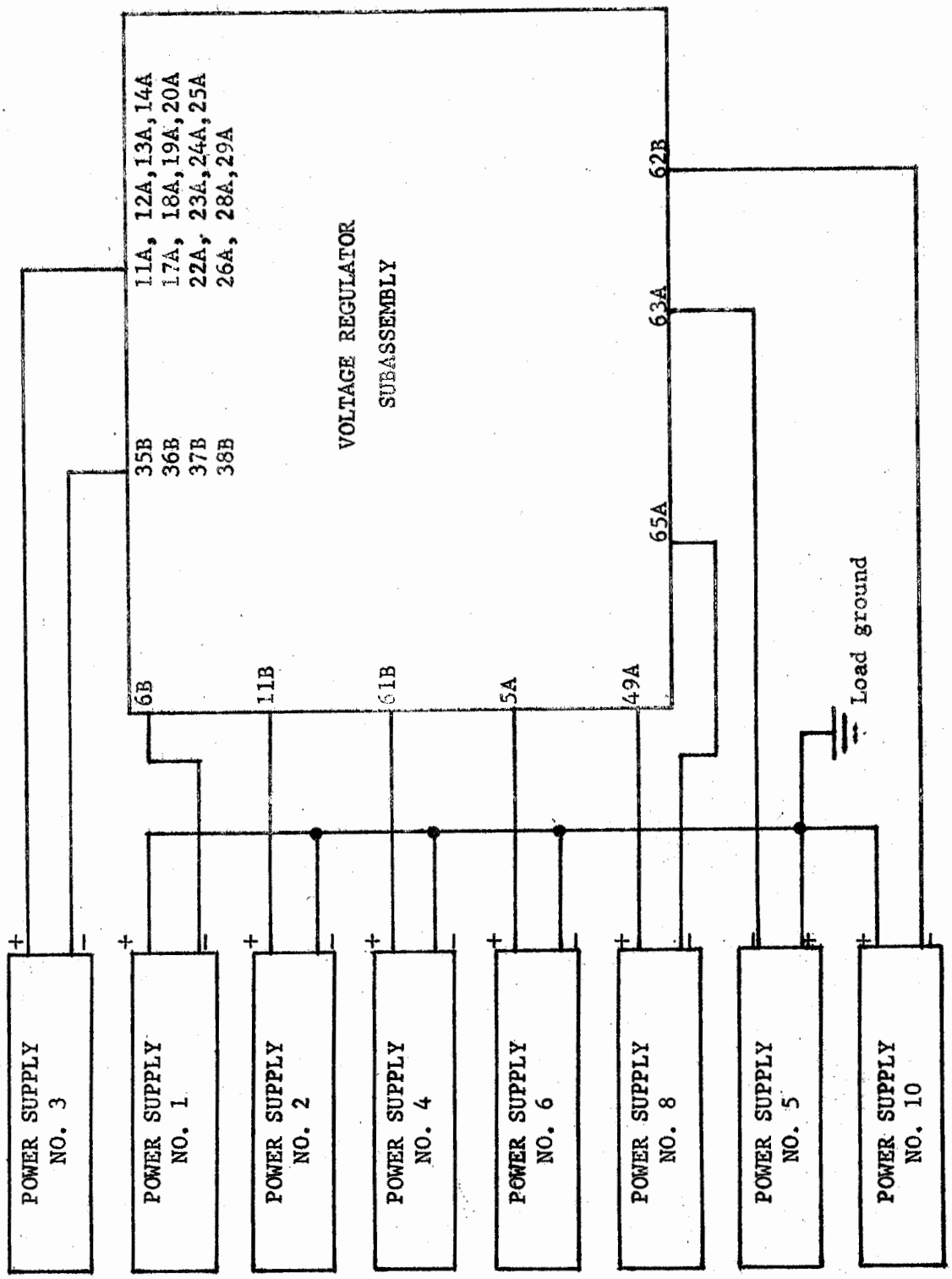



FIGURE NO. 1- Power Supply Connections for Voltage Regulator



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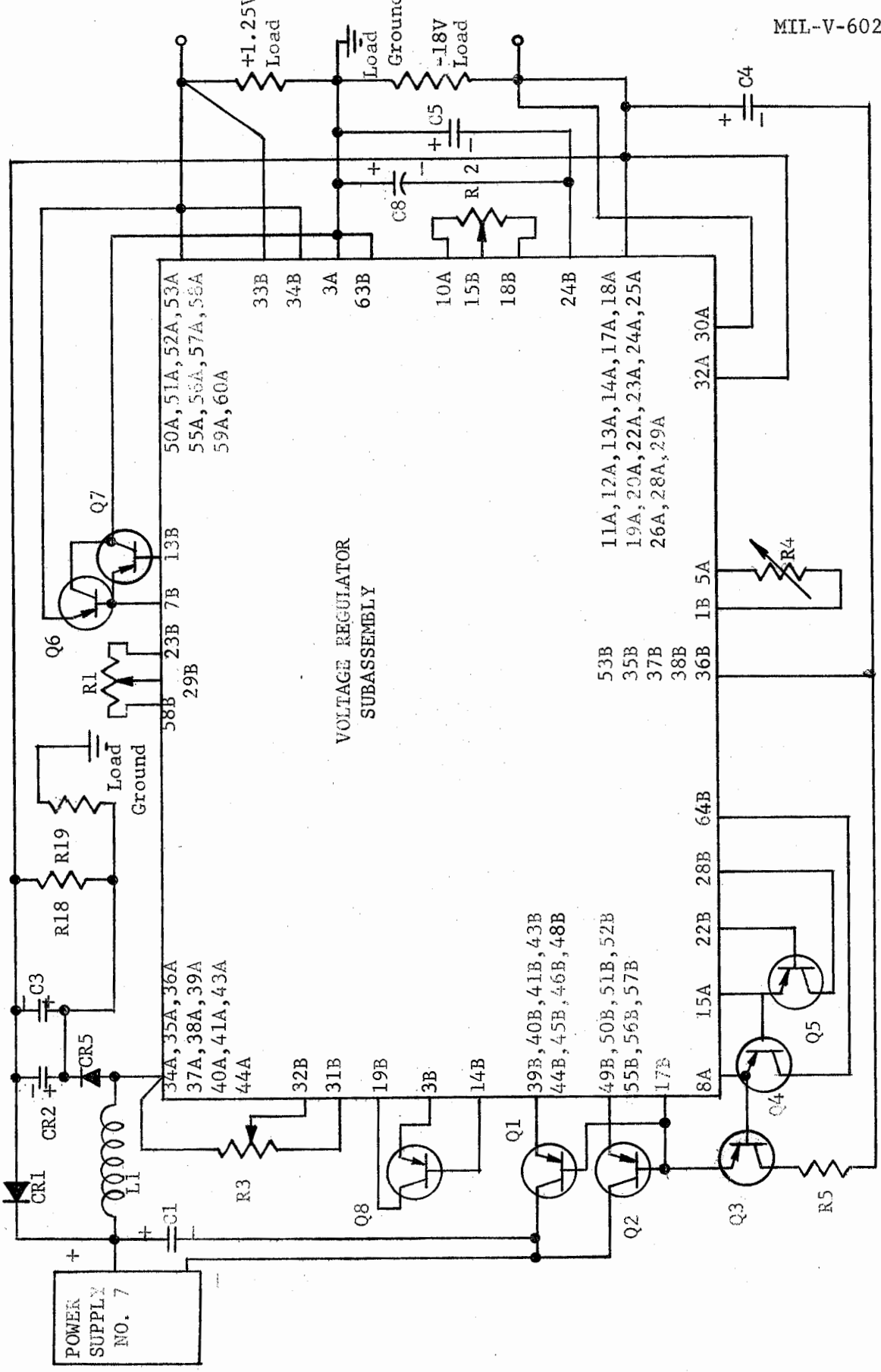



FIGURE 2 — External Circuitry for Voltage Regulator

NOTE: See Table No. VI and Table No. VII for part descriptions.



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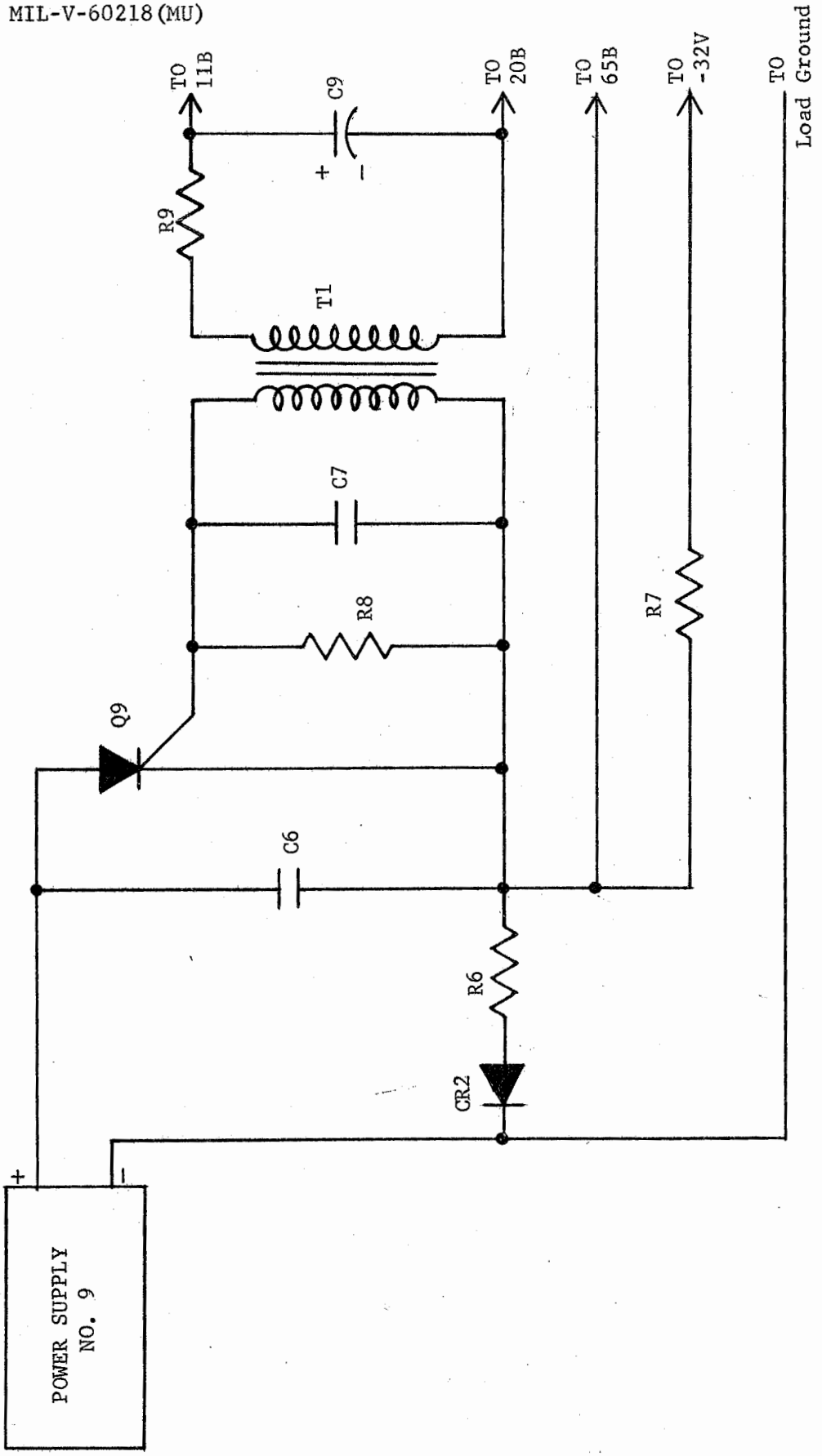



FIGURE NO. 3 - MASTER Trip Circuit for Voltage Regulator

NOTE: See Table No. V and Table No. VII for part descriptions.
Terminals refer to Voltage Regulator



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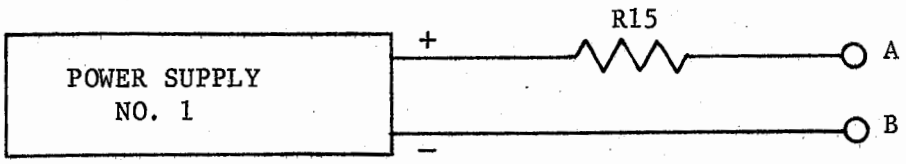



*FIGURE NO. 4 – Connections for Checking Thermistors

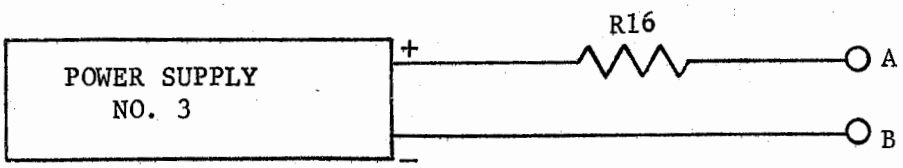



*FIGURE NO. 5 – Connections for Checking Resistors

*NOTE: See Table No. V and Table No. VII for Part Descriptions



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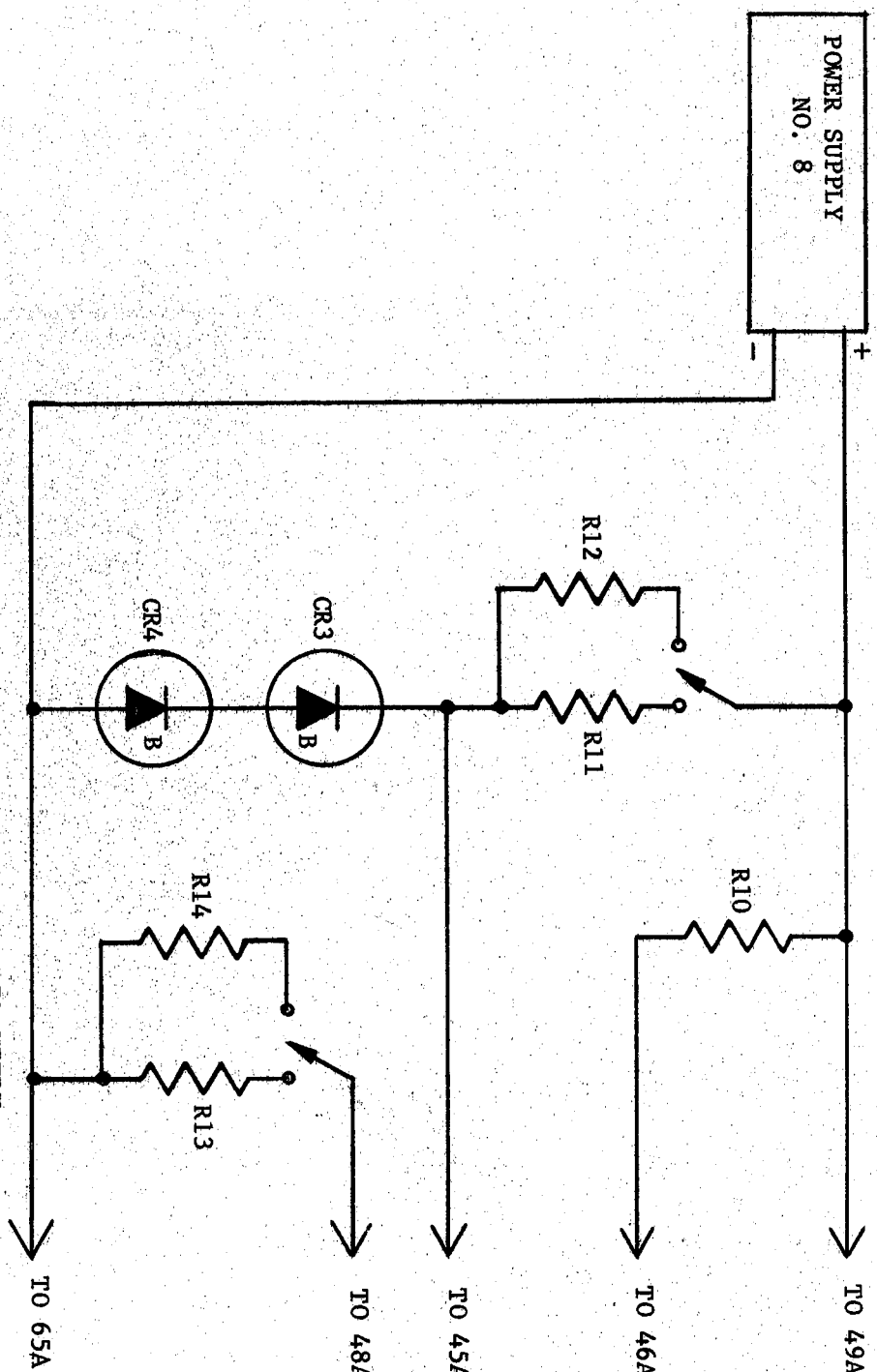



FIGURE 6 - NEON NIXIE SUPPLY EXTERNAL CIRCUITRY

NOTE: See Table No. VI and Table No. VII for Part Descriptions. Terminals refer to Voltage Regulator.

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